TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Analysis of Group IV (Ca, Ba, Sr)

The carbonate group — three white carbonates separated by chromate and sulphate solubilities and read finally by their flame colours: brick-red, crimson and apple-green.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 9 · LEVEL 1 TERM 1

1. Aim and the problem of three whites

Aim: to analyse the Group-IV precipitate — the white carbonates of Ca^{2+}, Sr^{2+} and Ba^{2+} thrown down by $(NH_4)_2CO_3$ (Experiment 4) — and identify which are present. All three precipitates are white and all three cations are colourless in solution, so this group is separated by fine differences in solubility and confirmed by the flame test, where the three finally show distinct faces.

2. The separation

1. Dissolve the washed carbonate precipitate in the minimum of dilute (ethanoic/acetic or dilute HCl) acid, as the laboratory scheme directs; boil off CO_2. The standard separation below uses the acetate solution.
2. **Remove barium first:** add potassium chromate (K_2CrO_4) solution: **$BaCrO_4$ precipitates yellow** — it is insoluble in acetic acid, while $SrCrO_4$ and $CaCrO_4$ stay dissolved under these conditions. Filter. Residue → barium, confirm by flame (apple green).
3. **Remove strontium next:** to the filtrate add ammonium sulphate (or dilute H_2SO_4 conditions per scheme): **$SrSO_4$ precipitates white** — $CaSO_4$ is appreciably more soluble and stays (in dilute solution). Filter. Residue → strontium, confirm by flame (crimson).
4. **Calcium remains** in the final filtrate: add ammonium oxalate: **white CaC_2O_4 precipitates**, insoluble in acetic acid but soluble in dilute HCl. Confirm by flame (brick red).

The order is forced by the solubilities: chromate falls for barium alone from acetate solution, and sulphate falls for strontium ahead of calcium only while solutions are dilute — reverse the order and the separations smear into each other.

3. Confirmatory summary

CATION	WET CONFIRMATION	FLAME

Ba^{2+}	K_2CrO_4 → yellow $BaCrO_4$, insoluble in acetic acid	Apple green
Sr^{2+}	$(NH_4)_2SO_4$ → white $SrSO_4$ (from the chromate-free filtrate)	Crimson (scarlet-red)
Ca^{2+}	Ammonium oxalate → white CaC_2O_4, insoluble in acetic acid, soluble in dilute HCl	Brick red

Flame technique decides these confirmations: the wire must be cleaned in concentrated HCl until it gives no colour, the salt taken up on the moistened wire (or as chloride solution), and the colour observed at the edge of the non-luminous flame the instant the sample enters — barium's green is brief and easily missed if the wire lingers.

4. Notes

- Traces of sodium (golden yellow, from reagents and glass) haunt every flame test in this group; judge the alkaline-earth colours by their first flash before sodium floods the flame.
- Calcium oxalate's insolubility in acetic acid distinguishes it from magnesium's oxalate behaviour should any Mg have leaked through — the reason NH_4Cl discipline in Experiment 4 matters here.
- Strontium and calcium sulphates differ in degree, not kind: work dilute, add reagent slowly, and let $SrSO_4$ stand to complete.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. State the order in which Ba, Sr and Ca are separated in Group IV and the reagent used at each step.
2. Why does K_2CrO_4 precipitate barium from the acetate solution but leave strontium and calcium?
3. Give the flame colours of the three group members and the wire-cleaning rule that makes them trustworthy.
4. How is calcium finally precipitated, and how does its precipitate behave in acetic acid and in dilute HCl?
5. A white Group-IV carbonate gives an apple-green flame flash. Name the cation and the wet confirmation you will still perform.
6. Why must solutions be kept dilute at the strontium sulphate step?

Answer key

1. 1) Ba^{2+} removed as yellow $BaCrO_4$ with K_2CrO_4. 2) Sr^{2+} removed as white $SrSO_4$ with ammonium sulphate. 3) Ca^{2+} precipitated from the remaining filtrate as white CaC_2O_4 with ammonium oxalate.
2. Because $BaCrO_4$ is insoluble even in acetic acid, whereas $SrCrO_4$ and $CaCrO_4$ remain soluble under the acetate conditions — a solubility gap wide enough, in acetic acid medium, to isolate barium alone.
3. Ba: apple green; Sr: crimson; Ca: brick red. Rule: clean the wire in concentrated HCl, heating repeatedly, until it imparts no colour to the flame before the sample is introduced.
4. With ammonium oxalate, as white calcium oxalate; it is insoluble in acetic acid but dissolves in dilute HCl — both halves of the behaviour are quoted in the confirmation.
5. Barium. Wet confirmation: dissolve the carbonate in dilute acetic acid and add K_2CrO_4 — a yellow precipitate of $BaCrO_4$ insoluble in acetic acid confirms it (the flame alone is indicative).
6. Because $CaSO_4$ is only moderately soluble: in concentrated solution it would precipitate with (or immediately after) $SrSO_4$ and the Sr/Ca separation would collapse; dilution keeps calcium’s sulphate below its precipitation threshold while strontium’s falls.
